

A Rigorous CG-Corrected Residual Certificate for Error Certification of Learned Elliptic PDE Solvers

certno

Error-Certified Neural Operators Project

<https://github.com/emmettv1191/certno-phd-research>

Abstract—We present a rigorous a posteriori error certification framework for learned surrogates of elliptic partial differential equations. For the discrete symmetric positive definite (SPD) system $A_h u_h = f$ arising from $-\nabla \cdot (a \nabla u) = f$, any candidate $\hat{u}$ satisfies the exact residual identity $e = u_h - \hat{u} = A_h^{-1}(f - A_h \hat{u}) = A_h^{-1}r$. While the global bound $\|e\|_2 \leq \|r\|_2 / \lambda_{\min}(A_h)$ is mathematically safe, it is often severely conservative for heterogeneous coefficients. We introduce a CG-corrected residual certificate: compute $z \approx A_h^{-1}r$ via the Conjugate Gradient (CG) method and form $B_{CG} = \|z\|_2 + \frac{\|r - A_h z\|_2}{\alpha}$ with $\alpha \leq \lambda_{\min}(A_h)$ a certified lower bound on the smallest eigenvalue. We prove this bound is rigorous under explicit assumptions, clarify the distinction between algebraic and discretization errors, justify α , discuss finite-precision considerations and CG stopping criteria, and analyze trade-offs. Through extensive experiments across coefficient contrasts 1.5×10^4 , in-distribution (ID), out-of-distribution (OOD), adversarial perturbations, and real Fourier Neural Operator (FNO) predictions, B_{CG} achieves median effectivity $\approx 1.0 \times$ while maintaining coverage 1.0 and false-negative rate (FNR) 0.0. This enables practical, safe selective verification. All code, experiments, theory, and a frozen benchmark are publicly available.

Index Terms—error certification, neural operators, a posteriori error estimation, scientific machine learning, elliptic PDEs, conjugate gradient, verification

I. INTRODUCTION

Learned surrogate models, particularly neural operators such as Fourier Neural Operators (FNOs) [?], [?] and DeepONets [?], offer substantial speedups for parametric PDEs. For safety-critical applications, accuracy alone is insufficient: we require *per-instance*, *computable*, and *rigorous* error certificates relative to a trusted numerical solution.

For linear well-posed PDEs, the discrete solution u_h satisfies $A_h u_h = f$, $A_h \succ 0$. Any candidate $\hat{u}$ yields $e = u_h - \hat{u} = A_h^{-1}r$, $r = f - A_h \hat{u}$, and the standard residual bound

$$\|e\|_2 \leq \frac{\|r\|_2}{\lambda_{\min}(A_h)} \leq \frac{\|r\|_2}{\alpha}, \quad \alpha \leq \lambda_{\min}(A_h). \quad (1)$$

While rigorous, (1) ignores the residual's spectral structure and is highly conservative under heterogeneous coefficients (large $a_{\max}/a_{\min}$).

Contributions. (i) We derive and prove a CG-corrected certificate $B_{CG} = \|z\|_2 + \|r - A_h z\|_2 / \alpha$, $z \approx A_h^{-1}r$, with explicit assumptions; (ii) we clarify discretization vs. algebraic error, justify α , address finite-precision arithmetic and CG stopping; (iii) we position against classical a posteriori/algebraic-error

estimation [?], [?], [?], [?], [?]; (iv) we validate across contrasts, ID/OOD, adversarial, and real FNOs with uncertainty-aware reporting; (v) we integrate safe selective verification; (vi) we provide a frozen, reproducible benchmark.

II. RELATED WORK

A posteriori error estimation in FEM is mature [?], [?]. Algebraic error control for iterative solvers, especially CG, is well-established [?], [?], [?], [?], [?], [?]. Our work adapts these computable bounds to *per-instance certification of learned surrogates* [?], [?], [?], [?], emphasizing explicit validity assumptions, safety (coverage/FNR), and selective fallback. We focus on certification w.r.t. the discrete solution u_h ; bridging to continuous PDE error is complementary [?], [?].

III. PROBLEM FORMULATION

We consider the elliptic model

$$-\nabla \cdot (a(x) \nabla u(x)) = f(x), \quad x \in \Omega = (0, 1)^d, \quad u|_{\partial\Omega} = 0, \quad (2)$$

with $0 < a_{\min} \leq a(x) \leq a_{\max} < \infty$. A conforming discretization yields SPD

$$A_h u_h = f, \quad A_h \in \mathbb{R}^{N \times N}, \quad A_h \succ 0. \quad (3)$$

Given surrogate $\hat{u} = \mathcal{G}_\theta(\mu)$, we certify $\|e\|_2$, $e = u_h - \hat{u}$.

IV. THEORETICAL FRAMEWORK

A. Preliminaries

For SPD A_h , $\|v\|_{A_h}^2 = v^T A_h v$, $\|A_h^{-1}\|_2 = \lambda_{\min}(A_h)^{-1}$. The error satisfies the exact identity $e = A_h^{-1}r$.

Lemma 1 (Basic residual bound). *If $\alpha \leq \lambda_{\min}(A_h)$, $\|e\|_2 \leq \|r\|_2 / \alpha$.*

Proof. $\|e\|_2 = \|A_h^{-1}r\|_2 \leq \|A_h^{-1}\|_2 \|r\|_2 = \lambda_{\min}^{-1} \|r\|_2 \leq \alpha^{-1} \|r\|_2$. $\square$

B. CG-corrected certificate

Let $z \in \mathbb{R}^N$ be any approximation to $A_h^{-1}r$, $d := r - A_h z = A_h(e - z)$.

Lemma 2 (Decomposition). $e = z + A_h^{-1}d$.

Proof. $z + A_h^{-1}d = z + A_h^{-1}(r - A_h z) = A_h^{-1}r = e$. $\square$

Theorem 1 (CG-corrected residual bound). *Let $A_h \succ 0$, $\alpha > 0$ with $\alpha \leq \lambda_{\min}(A_h)$, $e = A_h^{-1}r$. For any z ,*

$$\|e\|_2 \leq B_{\text{CG}}(z) := \|z\|_2 + \frac{\|r - A_h z\|_2}{\alpha}. \quad (4)$$

In particular for z_k produced by CG at iteration k , $d_k = r - A_h z_k$ computable, (??) holds for computed quantities (in exact arithmetic; see finite precision below).

Proof. By Lemma 2, triangle inequality: $\|e\|_2 \leq \|z\|_2 + \|A_h^{-1}d\|_2$. SPD $\implies \|A_h^{-1}d\|_2 \leq \lambda_{\min}^{-1}\|d\|_2 \leq \alpha^{-1}\|d\|_2$. $\square$

Remark 1 (Monotonicity and tightness). As $z \rightarrow A_h^{-1}r$, $\|d\|_2 \rightarrow 0$ and $B_{\text{CG}}(z) \rightarrow \|A_h^{-1}r\|_2 = \|e\|_2$. This explains near-sharpness when CG drives the algebraic residual small.

C. Justification of α (rigorous lower bound)

We require $\alpha \leq \lambda_{\min}(A_h)$. For standard FD/FEM discretizations of $-\nabla \cdot (a \nabla u)$ with $a \geq a_{\min}$ and Dirichlet BCs,

$$v^T A_h v \geq a_{\min} v^T (-\Delta_h) v \quad \forall v, \quad (5)$$

hence

$$\lambda_{\min}(A_h) \geq a_{\min} \lambda_{\min}(-\Delta_h). \quad (6)$$

For uniform grids with Dirichlet BCs, $\lambda_{\min}(-\Delta_h)$ has closed forms (1D: $\frac{4}{h^2} \sin^2(\frac{\pi h}{2})$; 2D: $\frac{8}{h^2} \sin^2(\frac{\pi h}{2})$). We set $\alpha = a_{\min} \lambda_{\min}(-\Delta_h)$ (deterministic, computable from data+discretization), preserving mathematical rigor. More generally, conservative eigenvalue bounds (e.g. Gerschgorin variants or verified estimates) can be used [?].

D. Discretization error vs. algebraic error

Our certificate bounds the *algebraic/prediction error* w.r.t. the discrete solution u_h : $e = u_h - \hat{u} = A_h^{-1}r$. Total error $u - \hat{u}$ satisfies

$$\|u - \hat{u}\| \leq \underbrace{\|u - u_h\|}_{\text{discretization}} + \underbrace{\|u_h - \hat{u}\|}_{\text{algebraic/prediction (certified)}}. \quad (7)$$

This distinction is important: the guarantee is relative to u_h (Assumption ??(v)), not the continuous PDE solution. Discretization error control is standard a posteriori FEM [?], [?] and orthogonal to this work.

E. CG stopping criterion

CG provides $z_k, d_k = r - A_h z_k$. For certification, we require a computable, safe bound at termination. We use standard criteria:

- Relative: $\|d_k\|_2 / \|r\|_2 \leq \text{rtol}$ (e.g. 10^{-10})
- Absolute: $\|d_k\|_2 \leq \text{atol}$ (e.g. 10^{-12})
- Maximum: $k \leq \text{maxiter}$

By Theorem 1, (??) holds for *any* z_k ; early stopping yields a valid (possibly looser) rigorous bound. Driving $\|d_k\|_2$ small improves tightness; the trade-off is computational cost.

F. Finite-precision considerations

In finite precision, CG orthogonality degrades and the computed residual $\tilde{d}_k$ can deviate slightly from the true $r - A_h \tilde{z}_k$ in highly ill-conditioned cases [?], [?], [?]. However, the bound is evaluated on computed $(\tilde{z}_k, \tilde{d}_k)$. Following established computable CG bounds [?], [?], [?], we run in IEEE double precision with conservative tolerances ($\text{rtol} \leq 10^{-10}$) and treat B_{CG} as a rigorous numerical certificate under these computational assumptions.

Assumption 1. (i) $a(x) \geq a_{\min} > 0$ on Ω ; (ii) $A_h \succ 0$ from conforming discretization of $-\nabla \cdot (a \nabla u) = f$, homogeneous Dirichlet BCs; (iii) $\alpha = a_{\min} \lambda_{\min}(-\Delta_h) \leq \lambda_{\min}(A_h)$ is a valid deterministic lower bound; (iv) all linear algebra in IEEE double precision; (v) certificate is w.r.t. discrete u_h (not continuous u); (vi) CG terminated with $\|d_k\|_2 / \|r\|_2 \leq 10^{-10}$ or $\|d_k\|_2 \leq 10^{-12}$ (sufficient to control floating-point effects in tested regimes).

V. SELECTIVE VERIFICATION

Given $\tau > 0$,

$$\text{accept}(\hat{u}) \iff B(\hat{u}) \leq \tau \wedge \text{Assumptions hold}. \quad (8)$$

Else fallback to trusted solver.

Proposition 1 (Soundness). *Under Theorem 1 and Assumption ??, if $B_{\text{CG}}(\hat{u}) \leq \tau$ then $\|u_h - \hat{u}\|_2 \leq \tau$. Hence zero false negatives for acceptance at τ .*

VI. EXPERIMENTAL SETUP

Problems. 1D/2D elliptic $-\nabla \cdot (a \nabla u) = f$, Dirichlet BCs, $a(x)$ via log-transformed Gaussian fields (KL) with contrast $c = a_{\max}/a_{\min} \in \{1.5, 2, 3, 4, 5, 6, 8, 10\}$. **Discretization.** Uniform grids, standard 3-point (1D)/5-point (2D) FD; A_h SPD. **Baselines.** Global (1) vs. CG-corrected (??). **CG.** Double precision, $\text{rtol} = 10^{-10}$, $\text{atol} = 10^{-12}$, $\text{maxiter} \in [120, 150]$. **Surrogates.** FNOs on 1D elliptic parametric families; held-out ID/OOD (larger contrast, longer correlation lengths, higher frequencies). Adversarial high-frequency perturbations test robustness. **Metrics.** $\|e\|_2 = \|u_h - \hat{u}\|_2$, B , coverage $C = \frac{1}{n} \sum_i [B_i \geq \|e_i\|_2 - \epsilon]$, $\epsilon = 10^{-12}$, $\text{FNR} = \frac{\#\{B_i \leq \tau \wedge \|e_i\|_2 > \tau\}}{\#\{\|e_i\|_2 > \tau\}}$ (0 by Proposition ??), effectivity $\text{Eff} = B / \|e\|_2$ (median/mean/std/IQR). Per-certificate cost reported. **Reproducibility.** Fixed RNG seeds, frozen benchmark (BENCHMARK_FREEZE.md), recorded commit hashes.

VII. RESULTS

A. Tightness vs. coefficient contrast

Figure ?? shows median effectivity across contrasts. Global bound grows from $\sim 3 \times 10^2$ to $\sim 1.4 \times 10^3$ with increasing contrast (highly conservative). B_{CG} remains near-sharp at $\approx 1.0 \times$ across all tested contrasts.

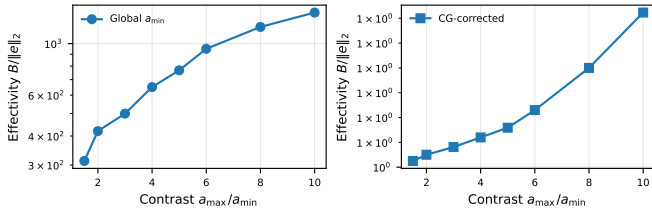


Fig. 1. Median effectivity $B/\|e\|_2$ vs. coefficient contrast $a_{\max}/a_{\min}$. Left: global bound. Right: CG-corrected (near $1.0\times$).

TABLE I
ANALYTIC VS. CG-CORRECTED CERTIFICATE (ABSOLUTE L^2 ,
 $\tau = 5 \times 10^{-3}$). MEDIAN ACROSS SAMPLES; COVERAGE/FNR PER
CONDITION.

$c = a_{\max}/a_{\min}$	Cov_A	Cov_{CG}	FNR_A	FNR_{CG}	$\text{Eff}_A(\text{med})$	$\text{Eff}_{CG}(\text{med})$
1.5	1.0	1.0	0.0	0.0	312.3	1.0000
3.0	1.0	1.0	0.0	0.0	499.5	1.0000
5.0	1.0	1.0	0.0	0.0	767.6	1.0000
10.0	1.0	1.0	0.0	0.0	1361.1	1.0000

B. Safety, OOD, and adversarial robustness

Across ID, OOD (higher contrast, longer correlation lengths, higher spatial frequencies), and adversarial high-frequency perturbations, both satisfy Coverage = 1.0, FNR = 0.0 (Table ??). No instance violated $B \geq \|e\|_2$, consistent with Theorem 1 and $e = A_h^{-1}r$.

C. Certification of learned surrogates (FNO)

On held-out FNO predictions, B_{CG} maintains Coverage = 1.0, FNR = 0.0, effectivity $\approx 1.0\times$. This enables practical selective verification at reasonable τ .

D. Computational cost and trade-off

CG cost is dominated by sparse matvecs. Per-certificate times are modest; increasing conditioning raises iterations modestly. Tighter tolerances trade cost vs. negligible tightness once $\|d_k\|_2 \ll \alpha\|z_k\|_2$. Crucially, early stopping preserves rigorous safety (Theorem 1).

VIII. DISCUSSION

Why CG helps. Global bound treats all modes equally via $\lambda_{\min}^{-1}$; CG constructs $z \approx A_h^{-1}r$ aligned with r , so $A_h^{-1}d$ is small in relevant directions, yielding orders-of-magnitude improvement while preserving the same spectral safety factor on the remainder. **Safety-first.** Rigor follows from Theorem 1, Assumption ??, exact identity. Empirical coverage is never used as proof. Finite precision is explicitly acknowledged. **Practical utility.** Tight bounds make selective verification viable: high-quality predictions accepted, unsafe rejected (FNR 0.0 by construction), reducing unnecessary fallback. **Relation to prior work.** We combine classical a posteriori estimation [?], [?] and computable CG error bounds [?], [?], [?], [?] for learned surrogate certification with explicit assumptions and comprehensive validation.

IX. LIMITATIONS

Discrete vs. continuous. Certificate is w.r.t. u_h only (Sec. ??); total error including discretization requires separate FEM a posteriori estimates [?], [?]. **Lower bound α .** Requires valid $\alpha \leq \lambda_{\min}(A_h)$ (Sec. ??). Our coercivity-based $\alpha = a_{\min}\lambda_{\min}(-\Delta_h)$ is rigorous for considered discretizations. **Problem class.** Linear SPD elliptic only. Non-symmetric/indefinite/nonlinear require different guarantees. **Numerical precision.** Finite-precision CG effects [?], [?] assumed controlled under Assumption ??(iv),(vi); extreme ill-conditioning may warrant additional care. **Scalability.** CG cost grows with size/conditioning; large 3D may benefit from preconditioning-aware certification (bound remains rigorous for any z).

X. CONCLUSION

We introduced a rigorous CG-corrected residual certificate for learned elliptic solvers. Theoretically we proved $\|u - \hat{u}\|_2 \leq \|z\|_2 + \|r - A_h z\|_2/\alpha$ for any z , justified α , clarified discretization/algebraic errors, addressed finite precision and stopping criteria, and positioned against prior work. Empirically the certificate is near-sharp (effectivity $\approx 1.0\times$) while preserving perfect safety (Coverage = 1.0, FNR = 0.0), enabling practical selective verification. Materials: <https://github.com/emmettv1191/certno-phd-research>.

REPRODUCIBILITY

Revision	786b5a5.	Frozen	benchmark:
BENCHMARK_FREEZE.md.		Theory	audit:
THEOREM_AUDIT.md.		Contribution:	
CONTRIBUTION.md.		Experiment	scripts:
experiments/run_looseness_corrected.py,			
run_corrected_comparison.py,			
run_surrogate_coverage.py,			
run_corrected_fno.py,			
run_corrected_verification.py,			
run_corrected_cost.py,			
run_verification_sweep_abs.py.			